



दक्षिण बिहार केन्द्रीय विश्वविद्यालय
CENTRAL UNIVERSITY OF SOUTH BIHAR

Question Paper of End-Term Examination, May 2026

Name of Department : Department of Computer Science Programme : M.Sc. (AI)
Session : 2025-2027 Semester : II
Course Title : Research Methodology Course Code : CAI82DC01104
Course Credit : 4 Max. Marks : 70 Duration : 3 hours
Enrolment No. (to be filled by Student) : CUS82502332006

Instructions: This question paper have three sections. The use of scientific calculator is permitted in this examination.

Section A

This section has ten questions of one point each. All questions are compulsory.

[1×10=10]

- Q1. What do you mean by term *exploratory* research?
- Q2. What are the steps in research process that should be completed before starting the development of the *Research Design*?
- Q3. Explain the term *coding* in data preparation process.
- Q4. What do you understand by the term *systemic bias in sampling*?
- Q5. Write the meaning of *extraneous variables*.
- Q6. Write the name of the statistical tools used in *Measures of dispersion*.
- Q7. Define the term *Modeling*.
- Q8. Compute the *Harmonic Mean* of the numbers 4, 5 and 10.
- Q9. What is purpose of footnotes in report writing?
- Q10. What is a *popular report*?

Section B

This section has six questions of six points each. Attempt any five questions.

[5×6=30]

Q1. Discuss the following

[3+3]

- I. Positive Skewness and Negative Skewness of the data set
- II. Coefficient of Skewness and its application

Q2. Ten competitors in a beauty contest are ranked by three judges in the following order:

I st judge	1	6	5	10	3	2	4	9	7	8
II nd judge	3	5	8	4	7	10	2	1	6	9
III rd judge	6	4	9	8	1	2	3	10	5	7

Use the Spearman's coefficient of rank correlation to determine which pair of judges has the nearest approach to common tastes in beauty,

Q3. Discuss the merits and demerits of interview method.

Q4. From a random sample of 36 New Delhi civil service personal, the mean age and the sample standard deviation were found to be 40 years and 4.5 years respectively. Construct a 95 percent confidence interval for the mean age of civil servant in New Delhi.

[The standard variate, z , for 95 percent confidence is 1.96, as per the normal curve area table]

Q5. Discuss the modeling process with a neat diagram.

Q6. Differentiate the following

[3+3]

- I. Descriptive vs. Analytical research
- II. Applied vs. Fundamental research

Section C

This section has three questions of fifteen points each. Attempt any two questions.

Q1. (a) Discuss the Hypothesis testing for variance.

(b) Solve the following question using χ^2 distribution.

A pharmaceutical company is considering the purchase of new bottling machines to increase efficiency. The factory currently makes use of machines that fill cough syrup bottles whose volume of medicine has a standard deviation of 1.6 mL. The new machine they are considering was tested in 30 bottles, producing a batch with a standard deviation of 1.2 mL. Does this machine produce a standard deviation less than 1.6 mL? Assuming normal distribution, test at 5% significance level. [Critical values tables are attached here for your reference].

[7+8]

Q2. Discuss the following informal research designs with the help of the neat and clean diagram. Also mention the limitation of each of the designs given below. [7.5+7.5]

I. Before-and-after without control design

II. After only with control design

Q3. Explain the term Interpretation in report writing. Also explain the different steps in writing the report?

[7.5+7.5]